

Q What happens at the first ultrasound?

The ultrasound at 8 to 14 weeks confirms your estimated delivery date, ensuring you get the right prenatal screening tests at the right time.

You may be asked to drink water before the ultrasound since a full bladder makes the ultrasound images clearer. During the ultrasound, the sonographer will determine the exact gestational age of your baby and

estimate your delivery date. At this stage all babies develop at the same rate, regardless of their future size. The measurements taken at this ultrasound are used to ensure that your baby is growing normally. Two key measurements are taken: the crown-rump length (the length from the top of the baby's head to the base of the spine), and the diameter from one bone across to the other on each side of your baby's head. The heart rate will also be measured and the position of the placenta will be checked.

This may be your first glimpse of your baby and it is a particularly exciting moment if she is moving around. You can ask to take a photograph from your ultrasound to share with others; you may have to pay for it.

First ultrasound This is the first of your routine ultrasounds. This will indicate how your baby is growing and developing.

Having the ultrasound The sonographer will move a small handheld transducer or probe across your skin to get a view of your baby (below).

White areas Hard tissues such as bone show up as white on the image.



Black areas These identify the fluids such as the amniotic fluid that the baby lies in.

Gray areas Soft tissues appear gray and speckled on the image.



Q What tests are available to check for abnormalities?

There are various tests that can be performed at different stages of your pregnancy to check for abnormalities. Some provide information on the risk of your baby being affected by an abnormality. These involve an ultrasound and blood tests. If these tests indicate that your risk is high, you will be offered further diagnostic tests, which allow specific conditions to be confirmed. Knowing for certain can help parents decide how to proceed, but these tests are invasive and do carry a 1 percent chance of miscarriage.

Q What is the combined test and when is it done?

To give you the most accurate risk calculation of various abnormalities, including Down syndrome, you should be offered the combined test, which involves a blood test and the nuchal translucency test. The blood samples are measured for levels of beta-human chorionic gonadotropin hormone (beta-hCG), which are usually higher in the blood of a mother carrying a baby with Down syndrome, and pregnancy-associated plasma protein (PAPP-A), levels of which are usually lower. The blood samples are usually sent for analysis, so it can take up to two weeks to receive the results. The blood samples must be taken between 10 and 14 weeks of pregnancy.

Q What are the quadruple and triple tests and when are they done?

You may be offered a quadruple test if you are between 14 and 20 weeks pregnant, when the combined test would no longer produce a reliable result. The quadruple test measures blood levels of beta-hCG, alpha fetoprotein (AFP; a protein produced by the fetus), unconjugated estriol (uE3: a form of estrogen that shows a lower level in babies with Down syndrome) and the hormone inhibin-A. The blood levels are analyzed in light of your age and how old the fetus is to produce a calculated risk. The triple- or Bart's—is the same as the quadruple test but does not look at levels of inhibin-A.